Mubazi: AI-assisted water meter reading and billing

Technical report, final product. Dushime Paulette, BSc Software Engineering (Machine Learning), African Leadership University. Supervisor: Samiratu Ntoshi.
Live app: wasac-smart-billing-qt9t.vercel.app · Code: github.com/Dushimepaulette1/wasac-smart-billing

1. Overview

WASAC water meters in Kigali are read by hand, which is slow and leads to estimated bills, misread digits and late discovery of leaks. Mubazi lets a household photograph its meter. Computer vision reads the eight wheels, the household confirms them, the reading is checked against that household's history, and a suspicious reading goes to a WASAC officer before any bill is made. The main objective in the proposal was a prototype that improves the accuracy, traceability and timeliness of readings. Because the reader is right on only 55.7% of unseen photos, the central design decision is that AI assists and people decide.

2. Scope alignment and feature validation

Planned feature	Intended purpose	Status	Evidence and justification
Local labelled dataset	Reflect real Kigali conditions	Completed	652 photos, 536 readable and 98 unreadable verified by eye; `notebooks/`
React web and mobile camera interface	Let households submit without an app store	Completed	Installable web app, live tips, English and Kinyarwanda; `frontend/src/pages`; Figure 2
FastAPI backend, PostgreSQL records	Traceable digital record of every reading	Completed	`backend/app`; deployed on Render and Neon
MobileNetV2 quality classifier	Catch unreadable photos	Completed, design changed	`app/vision/`. Runs on the server, not the device, and advises instead of blocking: fewer than 100 real unreadable photos, so wrongly blocking a household costs more than a bad photo
CRNN reader with CTC	Read the digits	Completed	`backend/ml/`. ResNet18 + BiGRU + CTC was compared with the plain CRNN and selected by a rule set before testing; Section 4
Isolation Forest and rules	Find implausible readings	Partial	`app/anomaly/`. Rules run live. The forest was evaluated on simulated data only (section 4) and is not deployed without real histories
Tariff bill calculator	Immediate bill estimate	Completed	`services/billing.py`. WASAC's published residential tariff with VAT on its own line, added after the video was recorded; older bills keep their own version. Other customer types not yet billed
Officer review queue	Person decides doubtful readings	Completed	Photo, previous reading, changed wheels, location; decisions audited; Figure 2, `pages/staff/ReviewQueue.tsx`
SMS through Africa's Talking	Receipts and alerts	Partial	`services/sms.py`. Built; sandbox only until a live account exists; staff outbox shows messages
USSD fallback	Serve basic phones	Partial	`services/ussd_menu.py`. Menu built and tested; a live channel needs an Africa's Talking account
MTN MoMo sandbox payment	Show the payment step	Completed	`services/payment_providers.py`. API flow tested against a fake MTN; no real sandbox settlement. Added MoMo Pay with staff confirmation for use without API keys
Three-month pilot, SUS, interviews	Evaluate with 10 to 15 households and 2 to 3 officers	Not implemented	No pilot ran: there was no formal WASAC partnership or access to real customers, so households stayed synthetic. No SUS scores were collected. End-to-end browser runs and a manual phone test stood in, which do not measure real usability

Proposal traceability. The supplied September proposal is mapped by phase, FR1–FR12 and NFR1–NFR7 in `docs/SCOPE_ALIGNMENT.md`. Its three-month pilot conflicts with the 12–28 November evaluation dates. Server-side advisory quality checks, review rather than automatic rejection, no deployed forest, and the absent officer field-round camera mode require an explicit scope decision. The original no-PII/no-retained-GPS/ALU-Drive-only plan differs from the contact/GPS records on Neon/Render; all current accounts are synthetic. No amended approval is claimed.

Added beyond the plan: YOLO to find the number window before reading; staff household upload; bill corrections with a second approver and account credit; next-bill estimates, possible-leak warnings and sector prioritization. Estimates use usual daily consumption; warnings are provisional, not proof of a leak. Sectors group by saved sector, summarize median monthly use and balances, and open the relevant account/review. Direct integration with WASAC's billing database stayed out of scope; customer records are synthetic and marked DEMO-.

3. Technical functionality

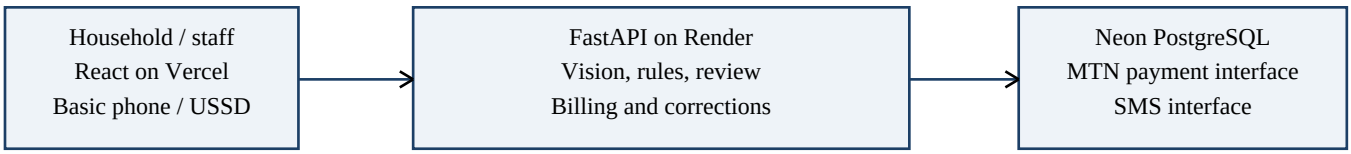


Figure 1. System architecture.

Architecture. A React and TypeScript single-page app (Vercel) calls a FastAPI service (Render, Docker) that serves three ONNX models with no PyTorch in the image, and stores data in PostgreSQL (Neon). Files are separated into routes, services, models, schemas, and `anomaly` and `vision` packages.

• **Reading.** `POST /submit-photo` runs YOLO, the quality gate and the reader. The household confirms the wheels on an editable counter.

• **Check first.** `POST /check-reading` runs the same read-only `evaluate_reading` as `/confirm-reading`, so the household learns whether a reading would be billed, held, or refused before it is stored. `/confirm-reading` repeats every check with the meter row locked.

• **Review and billing.** A reading below the last one, three or more times usual use, a first reading with no start, a value typed well below a reliable AI read, a rollover, or an odd location is held. An accepted reading creates a bill with the tariff version recorded.

• **Payment.** With MTN keys, MTN's status settles a payment, and an uncertain answer leaves it open rather than risking a second prompt. Otherwise the household dials WASAC's MoMo Pay code and gives the transaction ID, by app or USSD, and an officer confirms it against WASAC's merchant account.

• **Corrections.** Billed readings are never overwritten. Adjustments recalculate the bill and the next one; anything that raises a charge needs a second person.

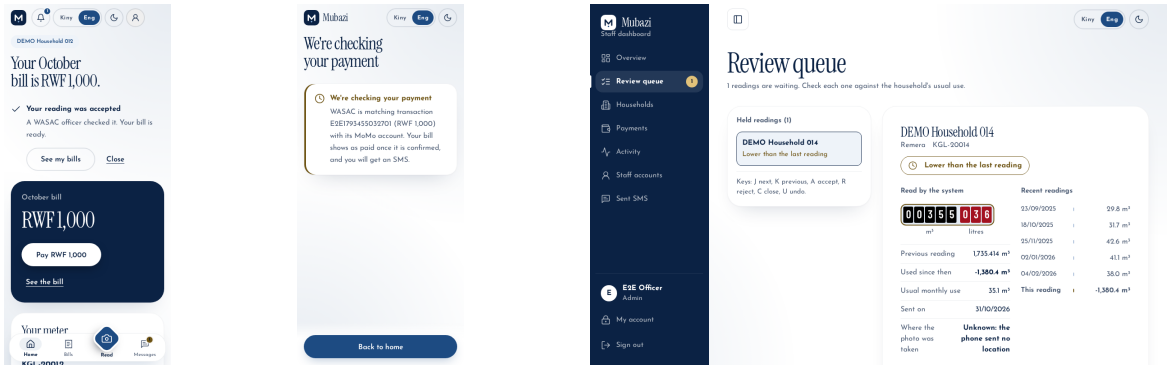


Figure 2. Household home with a bill; waiting for a MoMo Pay check; officer review with the evidence for a decision. From the automated end-to-end run with synthetic households.

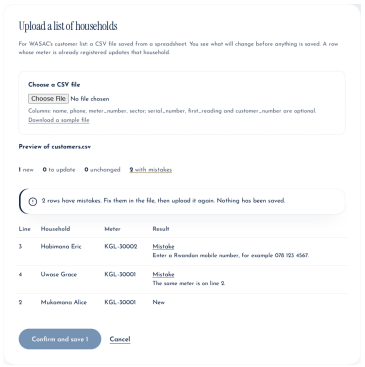


Figure 3. Upload preview with mistakes found: each row says what is wrong, and nothing can be saved until they are fixed.

4. Testing approaches and results

Strategy	Why, and how	Results and improvement made
Unit and integration	Money and rules must be exact. pytest and Vitest run on every push in GitHub Actions	Local run, 6 Nov: 617 backend passed, 5 skipped; 262 frontend passed (879 total). Coverage 94.10% backend and 83.36% frontend lines. Both macOS Python 3.9.6 and Linux production-image Python 3.10.22 passed; Node 22.23.4 and PostgreSQL 16. GitHub CI for the submitted commit remains required. Found the bugs listed under "What testing changed"
Concurrency	Two taps must not charge twice; SQLite cannot show this. Race tests on PostgreSQL 16, with a control that shows the race without the guards	16 local PostgreSQL checks passed, including fresh/legacy migrations. Row locks and partial indexes prevent duplicates; the negative-control race now uses a barrier instead of timing guesses
End-to-end	Proves the layers work together. Playwright drives a household and an officer through a real browser	6 Nov local run: 25/25 steps, zero problems, synthetic accounts and manual-entry fallback (no model weights). Acceptance, simulated payment and rejection passed. Historical live run was also 25/25
Model evaluation	Locked test set of 79 photos from 34 meters, split by meter, used once	55.7% reading accuracy (95% CI 42 to 69%), 70.9% right or within one digit. Led to confirmation of every reading by the household
Performance and stress	Free servers and slow networks. Local load test; deployed timings	Historical deployment measurements: production build on a phone screen, 194 KB, first paint 1.2 s on simulated fast 3G and 3.5 s on slow 3G, both with a 4x slower CPU. Local API: 0 errors at 50 concurrent users, median 143 ms for a bill. Deployed: 1.3 to 3.7 s per photo, about 1 minute after sleep. Peak memory 274 MB. Moving to ONNX Runtime made the API fit a 512 MB server. FPS does not apply: web pages have no graphics loop
Compatibility	Households use many phones. Chromium, Firefox and WebKit engines at four sizes	6 Nov: 36/36 checks passed in Linux Docker (Chromium 141, Firefox 142, WebKit 26, 360–1366 px), no page errors or horizontal scroll. Sizes and slow devices are simulated; Mac browser launches failed. Added to CI so every pull request is checked. A manual test on one phone over Wi-Fi found no problems; Android was not tried, and phone timings were not recorded.

Simulated anomaly evaluation. No real consumption histories exist, so detection was measured on a seeded simulation of 500 customers over 12 months (`backend/simulation/`), separate from the live system. Faults filled 5% of active months: leaks at 3 to 5 times normal use and meter bypasses at 0 to 5% of it. The forest trained without labels on 350 customers and was tested on 150 others (1,174 months, 88 of them faults). Intervals come from 2,000 bootstrap resamples of whole customers. **These results are synthetic, not evidence of real WASAC performance.**

Setup	Precision (95% CI)	Recall (95% CI)	F1 (95% CI)	F1, mean of 20 other seeds	F1, subtle faults
Rules only (live)	0.548 (0.325 to 0.775)	0.455 (0.271 to 0.640)	0.497 (0.311 to 0.657)	0.432	0.049
Forest only	0.854 (0.755 to 0.926)	0.932 (0.857 to 0.988)	0.891 (0.820 to 0.945)	0.884	0.667
Rules + Forest	0.689 (0.526 to 0.843)	0.932 (0.857 to 0.988)	0.792 (0.670 to 0.891)	0.809	0.600

The forest's F1 was 0.394 higher than the rules' (paired 95% CI 0.251 to 0.566). The rules caught none of the 36 bypass months, because the stuck-meter rule needs a true zero, while the forest caught 32. The rules also flagged 28.8% of normal commercial months, because the 5 m³ a day limit was set for households. These figures show that the method can work and where the rules are weak, not how it would perform in Kigali: the faults were designed by the author, and the forest's 5% contamination setting equals the injected rate, which real data would not reveal. The feature `consecutive_unpaid_months` was dropped, because one unpaid bill disconnects a simulated customer at once, so it is always 0 in the active months scored.

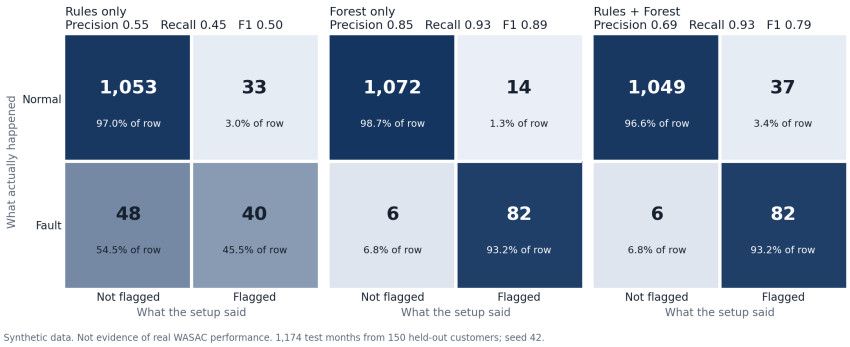


Figure 4. Confusion matrices on held-out simulated customers, seed 42. Synthetic data.

Different inputs. Tests cover a normal reading, a lower reading, three and ten times usual use, a first reading, meter rollover, repeats within 25 days, the same photo sent twice, wrong sign-in codes, malformed or reused transaction IDs, bad CSV rows, and MTN timeouts, 5xx and 404 answers. Each has an expected response; the table is in `docs/testing/README.md`.

What testing changed. Review found that an abandoned MoMo Pay attempt blocked bill corrections and held back credit, and that a top-up after a correction looked like a double payment. Each was reproduced with a failing test, then fixed. Testing also found that a meter added with no first reading billed the whole register, that readings at 3 to 10 times usual use were billed, and that blur checks fired more on readable photos, so they were turned off.

5. Code quality practices

• **Abstraction, inheritance and polymorphism.** `PaymentProvider` is an abstract class inherited by API, MoMo Pay and simulated versions, so `start_payment` never branches on a name. PyTorch and ONNX model classes meet the same `DigitReader` and `ReadabilityScorer` protocols, so the pipeline works with either.

• **Encapsulation.** `BillBalance` is the single definition of what is owed; dataclasses such as `AnomalyResult` and `Evaluation` hold results; each service has its own error class.

• **Separation of concerns and SOLID.** Routes only translate HTTP; rules live in services. `anomaly` is split into scoring (pure), evaluation (read-only) and service (writes). Adding a provider needs a new class, not edits to the payment flow.

• **Operations.** Alembic revisions preserve legacy rows and stop startup on failed schema updates. Photo cleanup runs at startup and hourly while awake, with a separate scheduler command for sleeping hosts. A local PostgreSQL dump/restore preserved 40 synthetic customers, 960 readings, the schema revision and test bill/payment amounts. Live Neon recovery and monitoring setup remain unverified.

• **Reuse and maintainability.** Typed API client, shared UI components, CSS design tokens, one read-only check for preview and submit. CI runs a strict type check. Naming, docstrings and folder layout are consistent.

• **Algorithms.** CTC decoding with digit confidence, per-household features, a reset-aware history, tier walking, oldest-first credit settlement, and row locks with idempotency keys. Each is linear or close to it in a small input. Costs and reasons are in `docs/ALGORITHMS.md`.

6. Limitations and next steps

Accuracy is moderate on a small dataset, and the interval is wide. Thresholds are provisional, and the tariff's effective date, service charge and other customer types need WASAC's confirmation. The anomaly model has no real data, and the billing rules in its simulation, including a 20,000 RWF penalty and disconnection after one unpaid bill, are assumptions; SMS and USSD need live accounts, and free hosting is slow after sleep. 56 Kinyarwanda placeholders await human translation; a review CSV and proposed pilot protocol are prepared. No SUS, real-user benefit, legal compliance, live delivery or 99% uptime result is claimed. Next steps are read-only access to WASAC's customer and meter records under a data-sharing agreement that complies with Law No. 058/2021, a real pilot, and retraining on those data.